LESSON PLAN · SCRATCH (GR 3–5) ADVANCED: ARCADE ENGINEERS

Lesson 2: Clones with Personality

Module 1 | 45 minutes | Clones + variables and sensing | Grades 3–5 · ages 9–11

Objective: Give clones different speeds, sizes and reactions, and detect when a clone touches the player.

Success criteria

- I can make smooth player controls.
- I can give clones random differences.
- I can detect when a clone touches the player.

Key vocabulary

- **collision**: When two sprites touch.
- **polling**: Checking something again and again in a loop.
- **randomise**: Give something a chance-based value.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Catch game (5 min): one learner holds a bin and moves along a line; others drop paper stars from random places; count catches.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Catch the falling stars: learners follow the steps.
14 min	**Perform**	Code Lab: smooth movement, Code Lab: catch the clones
9 min	**Probe**	Debug it, activity
4 min	**Wrap-up**	Exit ticket: Write the rule (if … then …) that scores a caught star.

Unplugged warm-up

Catch game (5 min): one learner holds a bin and moves along a line; others drop paper stars from random places; count catches.

Teaching the idea

Clones can be individualised with random values and can each sense collisions. Collision handling (score + delete) must sit inside the clone's movement loop. Smooth player movement uses polling (forever + if key pressed) rather than key events.

Analogy: Every raindrop follows the same rules but lands in a different place.

Live demo: Compare "when key pressed" movement (jerky) with forever + if key pressed (smooth) side by side.

Questions to ask

Ask	Listen for
Where must the touching check go?	Inside the clone's falling loop.
Why delete the clone after scoring?	Otherwise it scores again on the next frame.
How do we make clones different?	pick random for size, position or speed.

Differentiation

Support
Provide the smooth-movement script; learners add the touching rule.

Stretch
Add "bad" clones (a second costume) that cost a life when caught.

Common misconceptions

Misconception	How to fix it
Key-pressed hats give smooth movement	They trigger once per press; polling in forever is smooth.
The original sprite does the touching check for all clones	Each clone checks for itself in its own script.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Write the rule (if … then …) that scores a caught star.

Homework
Sketch a game where clones are enemies. What happens when one touches the player?

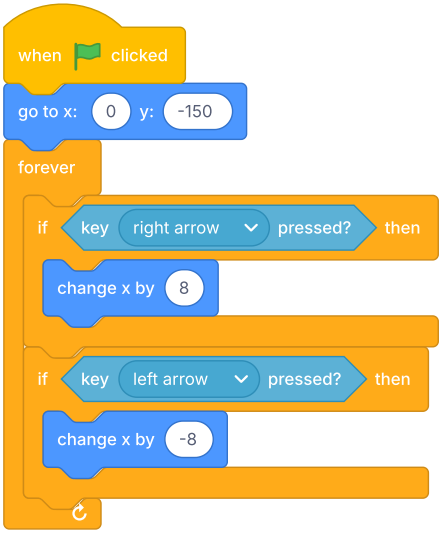
Answer key — Kitabash activities

1. Code Lab: smooth movement (perform · code lab)

Goals checked: Holding → glides right · Holding ← glides left · Checks keys in a forever loop

One solution:

Bowl

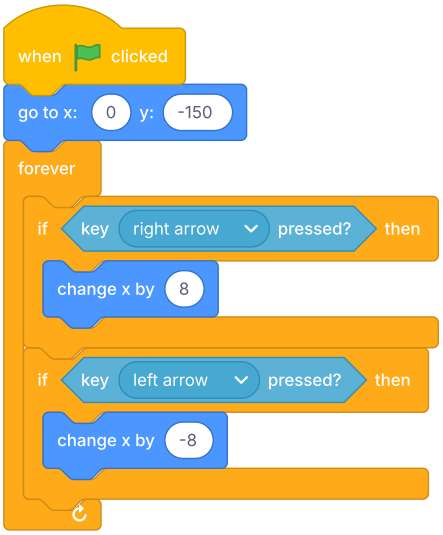


2. Code Lab: catch the clones (perform · code lab)

Goals checked: Clones fall · Catching scores · Score starts at 0

One solution:

Bowl



Star



3. Debug it (probe · multiple choice)

Answer: “delete this clone” after changing the score

4. Explain your thinking (probe · free response)

Look for: variable, speed, change. **Example:** I would make a variable called speed that starts at 3. Every time a star is caught, speed changes by 0.5, and each clone falls by speed, so later stars fall faster.

Answer key — worksheet

1. **Key words:** collision = When two sprites touch. · polling = Checking something again and again in a loop. · randomise = Give something a chance-based value.
2. **Plan your code:** Goals: Holding → glides right; Holding ← glides left; Checks keys in a forever loop.
3. **Plan your code:** Goals: Clones fall; Catching scores; Score starts at 0.
4. **Debug it:** A — The clone keeps touching the bowl and keeps scoring until it is deleted.

5. **Explain your thinking:** I would make a variable called speed that starts at 3. Every time a star is caught, speed changes by 0.5, and each clone falls by speed, so later stars fall faster.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.